System and Devices (SYS3)

Introduction to SYS3: Advanced Computer Systems

Dr Pengcheng Liu

COM00018I

SYS3 Team

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GTAs: William Carlton

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TEF Gold Teaching Excellence Framework

Staff roles in SYS3 Team

Dr Pengcheng Liu: Manage the module, give lectures, prepare and mark assessment.

Dr Christopher Crispin-Bailey: Design and supervise labs, prepare and mark assessment.

GTAs: Give assistance (not to give lectures). Ask them for help in labs!

What are we going to cover in the course?

Focus Areas of this Module

- Processor Design & Performance: Processor design, performance evaluation methods
- Instruction Pipeline: Instruction pipeline concepts, pipeline hazards
- Pipeline Scheduling: Static and dynamic scheduling, speculative execution, superscalar processors
- Memory: Virtual memory, page replacement, cache memory concepts and optimization techniques

What are we going to cover?

- Introduction: 0&1) Introduction to module & Advanced Computer Architecture 2) Introduction to Lab Practicals & Assessments
- Principles of Processor Design: 3) Review of Computer Organization Basics, 4) Performance Evaluation
- Instruction Pipeline Principles: 5) Introduction to RISC Pipeline, 6&7) Pipeline Hazards (Structural, Data and Control Hazards)
- Pipeline Scheduling: 8) ILP & Compiler, 9) Dynamic Scheduling, 10) Tomasulo's Algorithm 11) Speculative Execution 12) Superscalar processors
- Memory: 13) Virtual Memory, 14) Page Replacement, 15) Cache Memory Principles, 16) Block Replacement & Write Strategy, 17) Cache Optimization

Learning Outcomes

- Appreciate how, and why, the hardware structure of multiprocessor architectures differs from the architectures discussed previously. Use this knowledge to develop software applications for such architectures.
- Understand computer system design concepts to meet a performance requirement. Use this knowledge to evaluate the likely performance of a proposed computer architecture.
- Demonstrate the ability to optimise software for advanced hardware features like caches, branch predictors, and pipelines.
- Demonstrate through software development, how data structures are laid out in memory, how they can be accessed from a program, and how processor instructions are generated to manipulate them.

Lectures

- 2 lectures per week
  - pre-recorded
  - released a week before
- You will be required to do more reading as instructed
- Will not necessarily cover every detail of topics that you will be taught. Hence, a lot of independent learning is required
- Lectures
  - 17 lectures
  - 4 Tutorial Lectures at the end of each focus area

Lab Sessions

- 1 x 2 hour F2F practical sessions per week
- Split into 6 groups (you only need to attend the one scheduled for you)
- Check your timetable for time and room number of your group
- Lecturers and GTAs will be around to help
- Chris will explain the details in the lab practical briefing lecture

Lecture Review Sessions

- 2-hour Lecture Review sessions per week
- Whole Cohort
- Questions will be collected before the session and answered during the session
- Some of the sessions will be used to discuss the formative exercises and quizzes questions, please refer to the exercises and quizzes release schedule on VLE
- Agenda: Module Overview and Updates, Pre-Submitted Questions Discussion, SYS3 Exercises/Quizzes discussion (See related timeline on VLE), Discussion Forum, Closing Remarks and Announcements

Exercises and Assessments

- **Formative (not graded)**: 3 self-marked exercises. Please refer to the release schedule on VLE.
- **Formative (not graded)**: 3 auto-marked quizzes. Please refer to the release schedule on VLE.
- **Online Open Exam - (Centrally scheduled)**

  - Semester 2, Assessment Week

  - 3 hours

  - Weighting: 100%

VLE

All data for this module will be available via Blackboard:

- Announcements
- Lectures
- Lab instructions
- Exercises
- Quizzes
- Replay Content
- Padlet links for discussion forums and Lecture Review questions collection
- EBooks
- Other learning materials

Timetable

Please refer to online timetable:

- Reminder of the lectures, labs, lecture review sessions
- Data and time for the above

Suggested Reading Material

Main Book

- Computer Architecture: A Quantitative Approach, 5th Edition by Hennessy and Patterson (E-Book section on VLE Module page)

Suggested Books

- Operating System Concepts, Tenth Edition by Silberschatz, Galvin and Gagne. (Chapter of virtual memory)

Honor Code

- Indicate any assistance received on Assignments (friends, etc.).
- Do not copy solutions from other students or from the web.
- Do not give your solutions to others or post them on the web or any student forums.
- Report any inappropriate activity you see performed by others.
- Assignments are checked regularly for similarity with help of software tools.
- If you realize that you have made a mistake, you may retract your submission to any assignment at any time, no questions asked.
- If you need help, please contact us and we will help you.
- We do not want you to feel any pressure to violate the Honor Code in order to succeed in this course.

Golden Rules

- Come on time – please do not be late to the sessions
- Attend every lab and lectures that you are scheduled for
- Turn off mobile phones during the labs

Guidelines

- Questions sent via email will be answered collectively during the lecture review sessions
- Any other issues: email us or pop in the Office Hours: 13.00-14.00 at CSE/137, every Thursday, Semester 2 Week 1-11.

All the Best for the Module